COMP 551 Assignment 4 Group 2

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Abstract

We study text classification on the AG News dataset using two sequence modeling approaches: a word-level LSTM trained from scratch and a fine-tuned pretrained Transformer (DistilBERT). The LSTM requires explicit design choices for tokenization, vocabulary construction, and padding, while the Transformer leverages subword tokenization and pretrained representations. The LSTM achieves a test accuracy of 90.8%, while a bidirectional variant yields negligible improvement at higher computational cost. In contrast, DistilBERT reaches 94.4% test accuracy, demonstrating a substantial performance gain. Training dynamics show that the LSTM overfits after a few epochs, whereas the Transformer trains more stably. Error analysis reveals consistent confusion between semantically overlapping categories, particularly Sci/Tech and Business, suggesting that dataset ambiguity limits further gains. Overall, the results highlight the importance of pretraining and subword tokenization over architectural modifications in this task.

1 Introduction

Text classification is a core task in sequence modeling, where the objective is to assign labels to variable-length text inputs. In this project, we study this problem on the AG News dataset, a four-class classification task covering World, Sports, Business, and Sci/Tech. While the task is straightforward, performance depends heavily on how text is represented and how sequential information is processed.

We compare two distinct approaches. The first is a word-level LSTM trained from scratch, which requires constructing a vocabulary, handling unknown words, and managing padding explicitly. This makes the preprocessing pipeline transparent but limits the model to patterns learned solely from the training data. The second approach fine-tunes a pretrained Transformer (DistilBERT), which uses subword tokenization and representations learned from large external corpora, shifting complexity away from preprocessing and toward model scale.

A key difference lies in how text is tokenized and encoded. Word-level tokenization maps tokens to a fixed vocabulary and collapses rare words into a single unknown token, while subword tokenization decomposes words into smaller units and preserves more information. Similarly, LSTMs require explicit handling of padding when forming sequence representations, whereas Transformers use attention masks to ignore padded positions.

Our results show a clear performance gap between these approaches: the LSTM achieves around 90.8% test accuracy, while DistilBERT reaches 94.4%. This gap motivates a closer examination of whether improvements come from architectural choices, additional context, or pretrained representations, and how much performance is ultimately limited by ambiguity in the dataset itself.

2 Methods

2.1 Data and Split

We use the AG News dataset, which consists of 120,000 training examples and 7,600 test examples across four classes: World, Sports, Business, and Sci/Tech. The training data is split into 90% training and 10% validation using a fixed random seed for reproducibility, resulting in 108,000 training and 12,000 validation examples. All preprocessing steps, including vocabulary construction for the LSTM, are performed using only the training split.

2.2 Task 1: LSTM Classifier Trained from Scratch

Tokenization and vocabulary. We apply lowercasing and a basic word-level tokenizer. A vocabulary is constructed from the training data by keeping tokens with frequency greater than 2, resulting in a vocabulary of approximately 35k tokens. Special tokens include `PAD` and `UNK`. Each input is converted into a sequence of token IDs and truncated to a maximum length of 128.

Padding and sequence handling. Sequences are padded within each batch to a fixed length. To prevent padded tokens from affecting the representation, we use a length-aware approach based on the true sequence lengths.

Model architecture. The model consists of an embedding layer, a single-layer LSTM with hidden size 256, and a linear classifier. We use the final hidden state corresponding to the last non-padded token as the sequence representation. A bidirectional variant is also implemented with the same configuration except for bidirectionality.

Training setup. The model is trained using cross-entropy loss and the Adam optimizer with a learning rate of 10^{-3} and batch size 64 for 6 epochs. Gradient clipping is applied to stabilize training. We track training loss, validation loss, and validation accuracy at each epoch.

2.3 Task 2: Pretrained Transformer Fine-Tuning

Tokenization and inputs. We use the pretrained tokenizer associated with DistilBERT, which performs subword tokenization and adds special tokens. Inputs are truncated to a maximum length of 128 and converted into token IDs. Attention masks are used to distinguish real tokens from padding.

Model. We fine-tune the DistilBERT model with a sequence classification head. The full model is trained end-to-end.

Training setup. Training is performed using cross-entropy loss and the AdamW optimizer with a learning rate of 2×10^{-5} and batch size 16 for 3 epochs. We track training loss, validation loss, and validation accuracy.

2.4 Additional Experiment Beyond the Minimum

For the LSTM, we compare a unidirectional LSTM with a bidirectional LSTM under the same training setup to evaluate the effect of incorporating future context.

For the Transformer, we analyze the effect of pretrained representations through direct comparison with the LSTM models under the same dataset and evaluation setup. This isolates the impact of subword tokenization and large-scale pretraining on classification performance.

3 Results and Analysis

3.1 Overall Performance

Table 1 summarizes the main quantitative results on validation and test data. As highlighted in the table, DistilBERT significantly outperforms both LSTM variants in predictive accuracy. While the unidirectional LSTM achieves a respectable 90.8% test accuracy, upgrading to a bidirectional architecture did not yield meaningful improvements, despite requiring roughly twice the training time per epoch. Conversely, the pretrained Transformer, though possessing significantly more parameters and requiring longer computational time, achieved a test accuracy of 94.4%.

Table 1: Model Performance & Efficiency.

Model	Val.	Test	Params	Time
LSTM (Uni)	91.2%	90.8%	∼9.6M	∼3m
LSTM (Bi)	90.9%	90.5%	∼10.1M	∼6m
DistilBERT	94.7%	94.4%	∼66.9M	∼23m

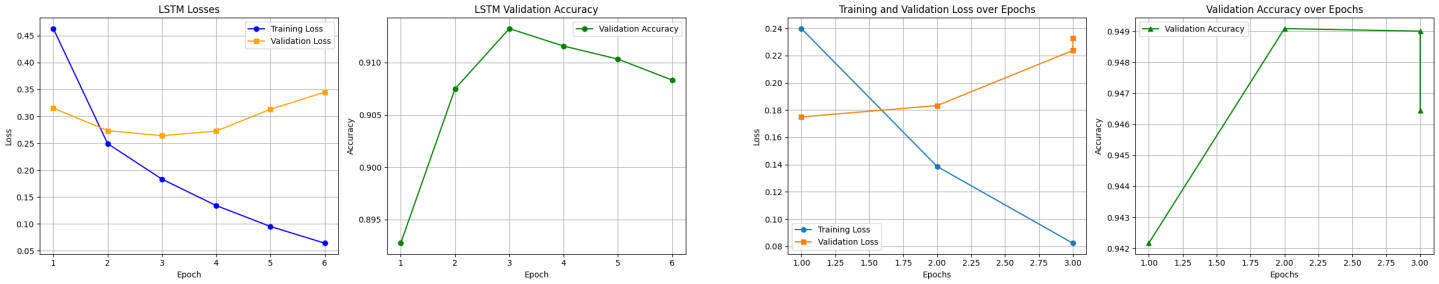
Table 2: DistilBERT PEFT Strategies.

Method	Val.	Test	Params	Time
Full Tune	94.9%	94.6%	66.9M	23:42
Frozen	90.2%	90.1%	593k	8:23
LoRA	93.2%	92.8%	741k	15:22

3.2 Training Curves

The training dynamics further highlight the differences between the two architectures. For the LSTM, the training loss continues to decrease sharply across all 6 epochs, but validation accuracy plateaus and validation loss begins to increase after epoch 3, indicating clear overfitting to the training vocabulary.

In contrast, DistilBERT demonstrates highly stable training. Leveraging its pretrained representations, the model rapidly converges within the first epoch and maintains a steady validation accuracy, showing robustness against overfitting despite its massive parameter count.



(a) LSTM Learning Curves

(b) DistilBERT Learning Curves

Figure 1: Training loss, validation loss, and validation accuracy over time.

See **Appendix B** for the comparison of learning curves between the model variations.

3.3 Qualitative Error Analysis

To understand the limitations of our models, we extracted misclassified examples from the test set. A recurring pattern emerged for both the LSTM and DistilBERT: heavy confusion between the **Business** and **Sci/Tech** categories.

Below are representative examples of misclassifications made by the DistilBERT model:

- **Example 1:** "*Microsoft releases new security patch for Windows Server...*"
  True Label: Sci/Tech — **Predicted:** Business
  Analysis: The presence of a corporate entity (Microsoft) likely triggered business-related attention weights, overriding the technical context of "security patch."
- **Example 2:** "*Oil prices surge as new extraction technology is deployed in the Gulf...*"
  True Label: Business — **Predicted:** Sci/Tech
  Analysis: The model fixated on "extraction technology," missing the broader macroeconomic subject of the article (oil prices).

This semantic overlap suggests that a portion of the error rate is constrained by the ambiguity of the dataset itself, where articles natively blend financial and technological reporting. See **Appendix A** for more example and **Appendix B** for a detailed confusion matrix of the models classifications.

3.4 Additional Experiment Analysis: PEFT and Architectural Changes

To go beyond the baseline requirements, we investigated how different training strategies and architectural modifications impact the accuracy-to-compute trade-off.

For the LSTM, we compared a unidirectional model with a Bidirectional LSTM to evaluate the effect of incorporating future context. As shown in Table 1, the Bi-LSTM provided negligible improvement (worsening slightly to 90.5% test accuracy) despite doubling the training time per epoch. This suggests that for short-text topic classification, sequential context is not the primary bottleneck.

For the Transformer, we explored Parameter-Efficient Fine-Tuning (PEFT) methods to reduce the computational burden of training DistilBERT's ∼66.9M parameters. We compared full fine-tuning against two alternatives: a Frozen Encoder (training only the classification head) and Low-Rank Adaptation (LoRA). The results are summarized in Table 2.

We observe a clear trade-off: the Frozen Encoder acts as a highly efficient feature extractor, training in a third of the time but sacrificing roughly 4% accuracy. LoRA offers a powerful middle ground, achieving near full-tuning accuracy (93.2%) while training a fraction of the parameters (∼1%). Full training curves for the LSTM variants and the model confusion matrices are provided in Appendix C for further reference.

4 Discussion and Conclusion

The main comparison in this project is between a scratch-trained recurrent model and a pretrained Transformer, which differ not only in architecture but in how text is represented and learned. The LSTM achieves strong baseline performance (90.8% test accuracy) with relatively few parameters, but exhibits clear overfitting and limited gains from architectural modifications. In particular, the bidirectional LSTM provides negligible improvement (worsening, in fact) despite roughly doubling training time, indicating that additional context alone does not resolve the main sources of error.

In contrast, DistilBERT achieves substantially higher accuracy (94.4%) and more stable training. This improvement is primarily due to subword tokenization and pretrained representations, which allow the model to handle rare words and capture richer semantics compared to the fixed vocabulary used by the LSTM.

From a practical perspective, the LSTM is simpler to implement and significantly smaller (millions vs tens of millions of parameters), but requires careful handling of tokenization, padding, and sequence lengths. DistilBERT simplifies preprocessing but is more memory-intensive and relies on a larger pretrained model. Despite this, its faster convergence and higher accuracy make it more effective overall for this task.

Error analysis shows that both models struggle with similar cases, especially between Sci/Tech and Business, where articles contain overlapping themes. This suggests that a portion of the remaining errors is due to dataset ambiguity rather than model limitations.

Overall, the results indicate that pretraining and subword tokenization have a far larger impact on performance than increasing recurrent model capacity. Furthermore, our exploration into parameter-efficient fine-tuning (PEFT) highlights a critical practical takeaway: full fine-tuning of massive models is not always strictly necessary. While freezing the entire encoder saves the most time, it sacrifices too much accuracy. However, injecting low-rank adaptations (LoRA) into the attention mechanisms provides an optimal middle ground—achieving near state-of-the-art performance by updating less than 1% of the parameters. This demonstrates that PEFT methods are essential for balancing high accuracy with the resource constraints of real-world deployments.

5 Statement of Contributions

- **Barry:** Focused on the bonus part of the assignment. Edited, debugged, and finalized Python notebook; Led the writing of the report.
- **Maeve:** Implemented Task 2 (DistilBERT Transformer), including subword tokenization, attention mask formatting, Hugging Face Trainer integration, and metric logging; co-wrote and edited the report.
- **Nick:** Implemented Task 1 (LSTM Classifier): pre-processing, tokenizing, training / validation, analyzing / visualizing performance; co-wrote and edited the report.

6 Statement on the Use of LLMs and Other Resources

We used large language model (LLM) tools as writing and development assistants in a limited manner. Specifically, we used ChatGPT to improve grammar, clarity, and academic tone, and to implement LaTeX layout and formatting adjustments. In all cases, we first drafted the report ourselves and then used these tools to refine wording and presentation; the technical content, experimental design, results, and interpretations remain our own. We reviewed and edited all AI-suggested changes to ensure correctness and consistency with our work.

For programming support, LLMs were used for debugging assistance and for generating plotting and visualization code. The core machine learning architectures, specifically the LSTM and Transformer classifiers (including preprocessing and training steps) were designed, engineered, and trained by our team.

References

[1] X. Zhang, J. Zhao, and Y. LeCun, "Character-level Convolutional Networks for Text Classification," *Advances in Neural Information Processing Systems*, 2015. Dataset version distributed via Hugging Face *ag_news*.

[2] PyTorch Documentation, "`torch.nn.LSTM`," `https://pytorch.org/docs/stable/generated/torch.nn.LSTM.html`.

[3] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *NeurIPS EMC2 Workshop*, 2019.

[4] Hugging Face Documentation, "Text classification," `https://huggingface.co/docs/transformers/tasks/sequence_classification`.

Appendix

A Extended Misclassified Examples

Below is the extended list of misclassified examples from the test set for both the DistilBERT and LSTM models, illustrating the recurring semantic overlap between categories (particularly Business and Sci/Tech).

A.1 LSTM Misclassifications

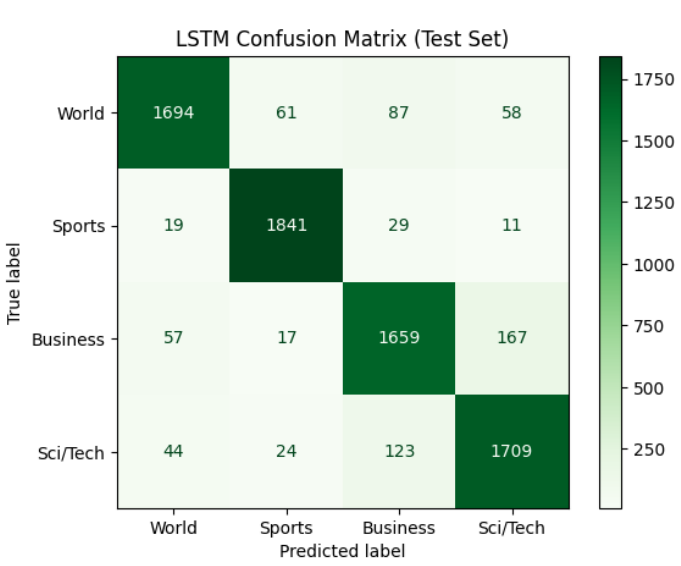
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Card fraud unit nets 36,000 cards In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests - and estimates it saved 65m.*
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Super ant colony hits Australia A giant 100km colony of ants which has been discovered in Melbourne, Australia, could threaten local insect species.*
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Rivals Try to Turn Tables on Charles Schwab By MICHAEL LIEDTKE SAN FRANCISCO (AP) – With its low prices and iconoclastic attitude, discount stock broker Charles Schwab Corp. (SCH) represented an annoying stone in Wall Street's wing-tipped shoes for decades...*
- **True:** World — **Pred:** Business — **Text:** *Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil market.*
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it.*
- **True:** World — **Pred:** Sports — **Text:** *Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.*
- **True:** Business — **Pred:** Sci/Tech — **Text:** *Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.*
- **True:** World — **Pred:** Business — **Text:** *Venezuela vote keeps Chavez as president CARACAS – Venezuelans voted resoundingly to keep firebrand populist Hugo Chavez as their president in a victory that drew noisy reactions yesterday from both sides in the streets. International observers certified the results as clean and accurate.*
- **True:** Business — **Pred:** World — **Text:** *Surviving Biotech's Downturns Charly Travers offers advice on withstanding the volatility of the biotech sector.*
- **True:** World — **Pred:** Sports — **Text:** *Mr Downer shoots his mouth off Just what Alexander Downer was thinking when he declared on radio last Friday that "they could fire a missile from North Korea to Sydney" is unclear. The provocative remark, just days before his arrival yesterday on his second visit to the North Korean ...*

A.2 DistilBERT Misclassifications

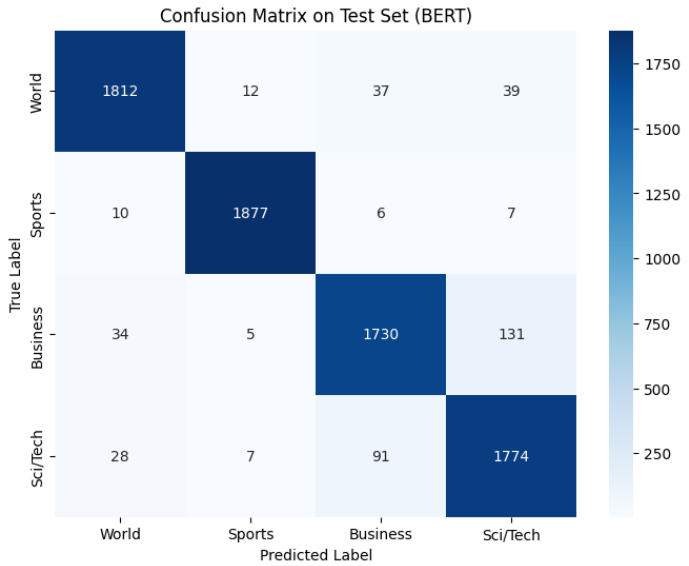
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual stranglehold it's had on IPOs. Public bi...*
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it...*
- **True:** Business — **Pred:** Sci/Tech — **Text:** *Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planne...*
- **True:** Business — **Pred:** Sci/Tech — **Text:** *Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space -...*
- **True:** Sci/Tech — **Pred:** Business — **Text:** *IBM Buys Two Danish Services Firms IBM said Tuesday it has acquired a pair of Danish IT services firms as part of its effort to broaden its presence in Scandinavia. As a result of the moves, IBM will ...*
- **True:** World — **Pred:** Business — **Text:** *Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed...*
- **True:** World — **Pred:** Sci/Tech — **Text:** *Bush Promotes His Plan for Missile Defense System President Bush, in Pennsylvania, said that opponents of a missile defense system were putting the nation's security at risk...*
- **True:** Sports — **Pred:** World — **Text:** *St. Louis Cardinals News Right-hander Matt Morris threw seven solid innings, but the Cardinals needed a bases-loaded walk to second baseman Tony Womack and a grand slam from new right fielder Larry Wa...*
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Small computers can have multiple personalities, too BOSTON The jury is still out on whether a computer can ever truly be intelligent, but there is no question that it can have multiple personalities...*
- **True:** Sci/Tech — **Pred:** Business — **Text:** *Intuit Posts Wider Loss After Charge (Reuters) Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a wider quarterly loss after taking a goo...*

B Supplementary Figures and Diagnostics

This section contains supplementary visualizations generated during our extended experiments, including the training dynamics of the LSTM variants and the confusion matrices used for our qualitative error analysis.



(a) LSTM (Unidirectional)



(b) DistilBERT (Full Fine-Tuning)

Figure 2: Confusion Matrices on the Test Set. Both models exhibit significant overlap between the Sci/Tech and Business categories, though DistilBERT correctly resolves many of the ambiguities that confused the LSTM.

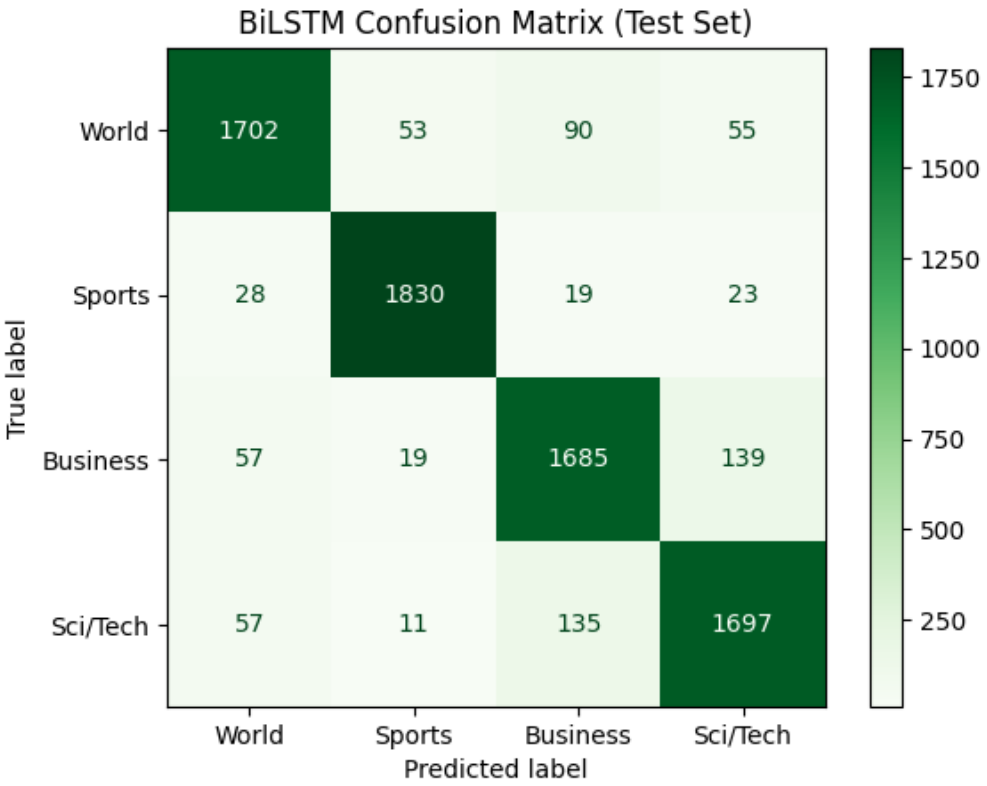


Figure 3: Confusion Matrix for the Bidirectional LSTM on the Test Set. Similar to the unidirectional model, the Bi-LSTM continues to struggle with the semantic overlap between the Sci/Tech and Business categories, demonstrating that adding future context did not resolve this core ambiguity.

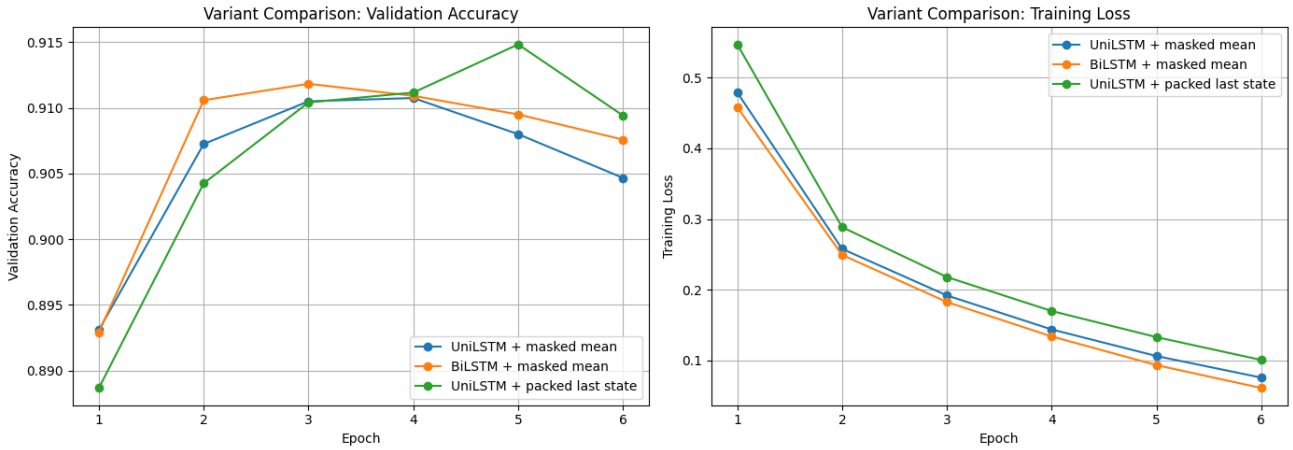


Figure 4: Validation Accuracy and Training Loss comparisons across the different LSTM variants (Unidirectional vs. Bidirectional). Both models exhibit similar overfitting trajectories after epoch 3.

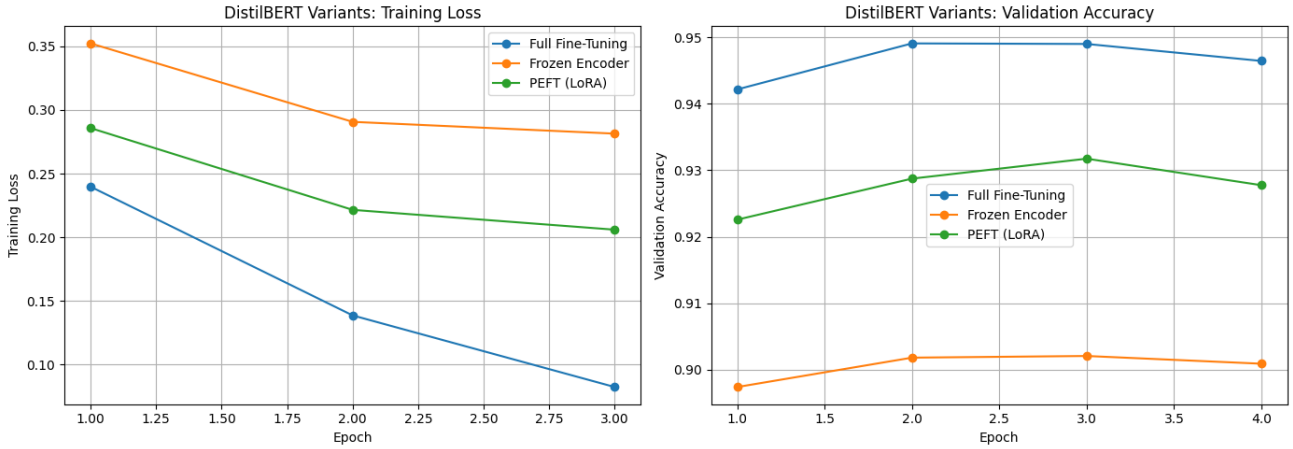


Figure 5: Validation Accuracy and Training Loss comparisons across the different BERT variants (Full, Frozen, PEFT with LoRA). All models exhibit the same training curve up to translation in term of accuracy.